ESERCIZIO POL1 NMOS

CALCOLARE i_D DI NMOSFET SAPENDO CHE

$v_{DS} = 0{,}25V$, $v_{GS} = 0{,}1V, 1V, 2V, 3V$

$W = 5\mu m$, $L = 0{,}5\mu m$, $K'_n = 200\mu A/V^2$

$V_{TN} = 0{,}8V$

[0, 40μA, 538μA, 1,04 mA]

ESERCIZIO POL2 NMOS

SI CONSIDERI MOSFET CON I SEGUENTI PARAMETRI

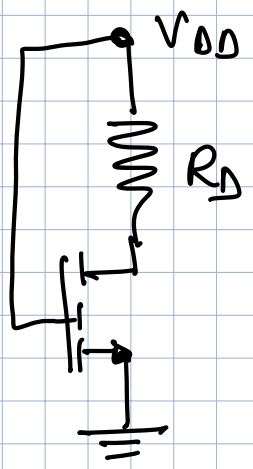
$K_n = 250\mu A/V^2$; $V_{TN} = 0{,}75V$; $\lambda = 0{,}025\ V^{-1}$;

$v_{GS} = 5V$; $v_{DS} = 6V$

1) CALCOLARE i_D
2) CALCOLARE i_D CON $\lambda = 0$

[2,60 mA ; 2,26 mA]

ESERCIZIO POL3 NMOS



$V_{DD_1} = 4V$; $V_{DD_2} = 5V$

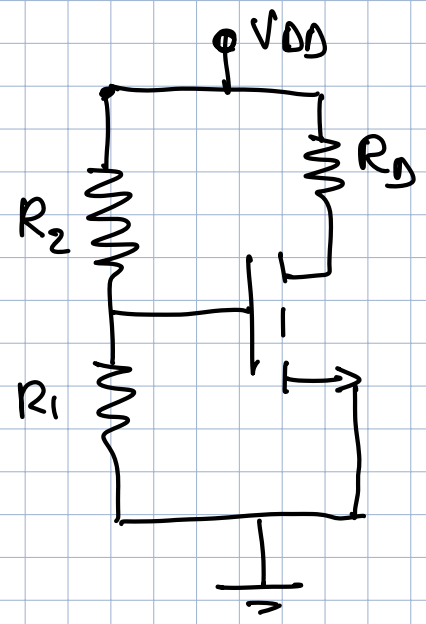
$R_D = 6{,}8k\Omega$

USANDO LA CARATTERISTICA DI USCITA DI P4.18 (ANCHE SU MOODLE) TROVARE IL PUNTO DI LAVORO

[$V_{DS} \cong 1{,}7V$, $i_{DS} = 350\mu A$]

[$V_{DS} \cong 1{,}5V$, $i_{DS} = 520\mu A$]

ESERCIZIO POL4 NMOS



$V_{DD} = 10V$ $\quad$ $V_{TN} = 2V$

$R_2 = 10 M\Omega$ $\quad$ $K_n = 0{,}25\ mA/V^2$

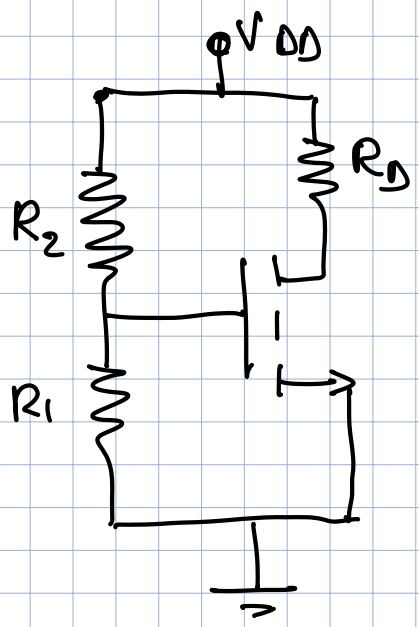
$R_1 = 10 M\Omega$

$R_D = 6 K\Omega$

TROVARE i_D ; V_{DS}

[$i_D = 1{,}125\ mA$, $V_{DS} = 3{,}25V$]

ESERCIZIO POL5 NMOS



$V_{DD} = 10V$

$R_1 = 750\ K\Omega$ $\quad$ $V_{TN} = 1V$

$R_2 = 270\ K\Omega$ $\quad$ $K_n = 25\ \mu A/V^2$

$R_D = 100\ K\Omega$

TROVARE i_D ; V_{DS}

[$i_D = 93{,}62\ \mu A$; $V_{DS} = 0{,}62V$]

ESERCIZIO POL6 NMOS

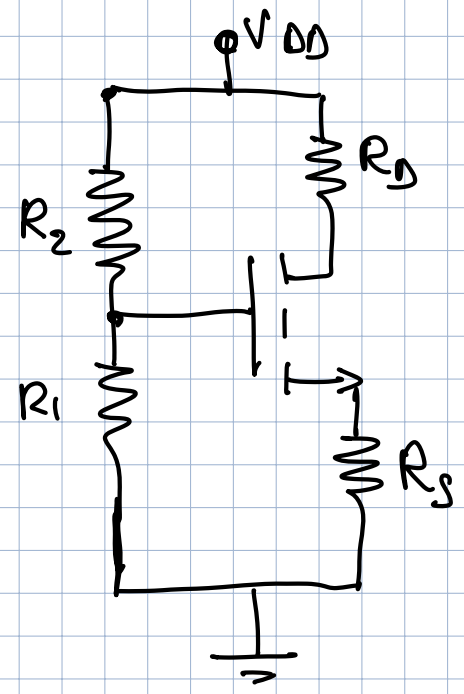
RIPETERE IL PROBLEMA POL5NMOS CON I NUOVI DATI: $R_1 = 300\ K\Omega$; $R_2 = 700\ K\Omega$

GLI ALTRI SONO UGUALI.

<u>Trovare i_D, V_{DS}</u>

[$i_D = 50\ \mu A$; $V_{DS} = 5V$]

ESERCIZIO POL7 NMOS



$R_1 = 100\ K\Omega$ $\quad V_{DD} = 16V$

$R_2 = 60\ K\Omega$ $\quad V_{TN} = 6V$

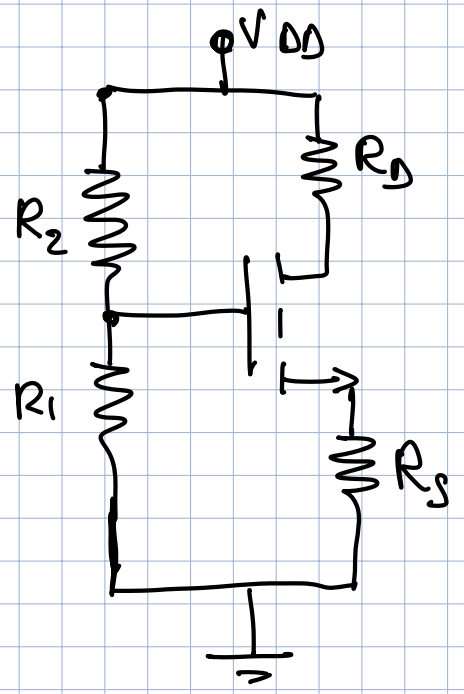
$R_D = 1{,}6\ K\Omega$ $\quad K_n = 444{,}4\ \mu A/V^2$

$R_S = 0{,}4\ K\Omega$

trovare i_D, v_{DS}

[$i_D = 2{,}18\ mA$, $v_{DS} = 11{,}64V$
$v_{GS} = 9{,}13V$

ESERCIZIO POL8 NMOS



$R_1 = 10\ M\Omega$ $\quad V_{DD} = 10V$

$R_2 = 10\ M\Omega$ $\quad V_{TN} = 1V$

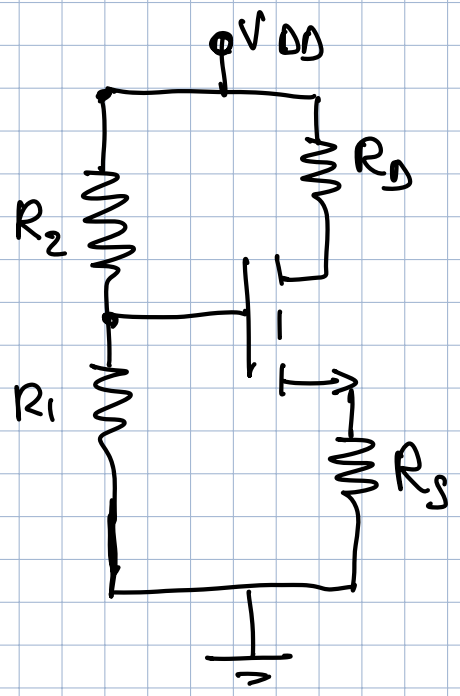
$R_D = 6\ K\Omega$ $\quad K_n = 1\ mA/V^2$

$R_S = 6\ K\Omega$

trovare i_D, v_{DS}

[$i_D = 0{,}5\ mA$, $v_{DS} = 4V$
$v_{GS} = 2V$

ESERCIZIO POLIN MOS



$V_{DD} = 15V$

$R_1 = 240\ K\Omega$ $\quad V_{TN} = 3V$

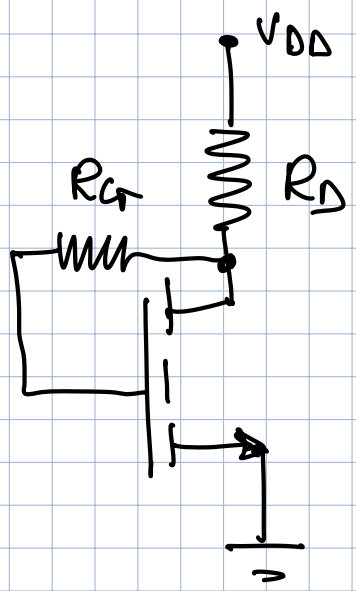
$R_2 = 240\ K\Omega$ $\quad K_n = 4\ mA/V^2$

Determinare R_D e R_S in modo che $i_D = 2mA$, $v_{DS} = 7V$

[$R_S = 1{,}75\ K\Omega$; $R_D = 2{,}25\ K\Omega$]

HP SATURA $\rightarrow$ $i_D = \frac{K_n}{2}(v_{GS} - V_{TN})^2$

ESERCIZIO POLIONMOS



$V_{DD} = 16V$ $\quad V_{TN} = 6V$

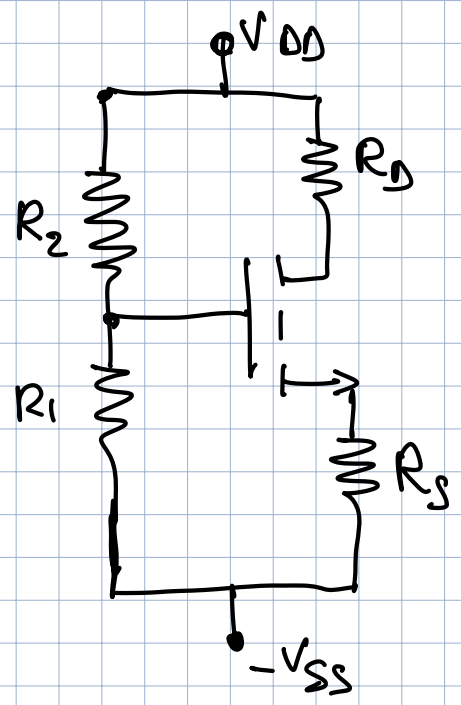
$R_G = 40\ K\Omega$ $\quad K_n = 444{,}4\ \mu A/V^2$

$R_D = 1{,}6\ K\Omega$

CALCOLARE i_D e v_{DS}

[$i_D = 3{,}70\ mA$; $v_{DS} = 10{,}08\ V$]

ESERCIZIO POL11 NMOS



$V_{DD} = 15V$ $V_{SS} = 15V$

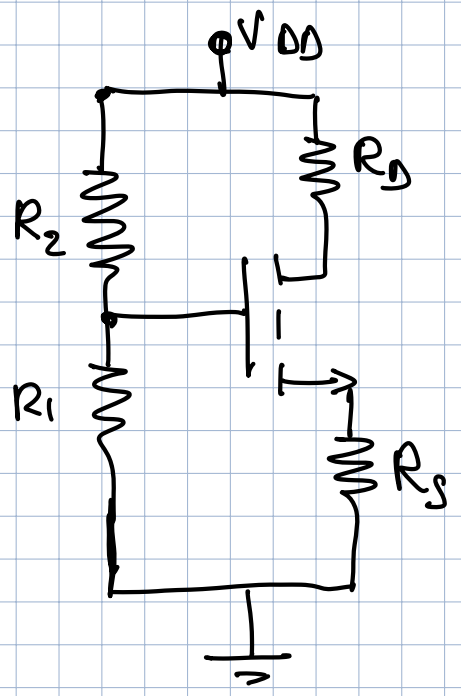
$R_1 = 600\ K\Omega$ $V_{TN} = 3V$

$R_2 = 400\ K\Omega$ $K_n = 10\ mA/V^2$

Determinare R_D e R_S in modo che $I_D = 20\ mA$, $V_{DS} = 3V$

[$R_S = 650\Omega$; $R_D = 700\ K\Omega$]

ESERCIZIO POL12NMOS



$V_{DD} = 12V$

$V_{TN} = 2V$

$K_n = 50\ \mu A/V^2$

Determinare le resistenze in modo che $I_D = 100\ \mu A$, $V_{DS} = 4V$

IN QUESTO ESERCIZIO SI HA PIÙ GRADI DI LIBERTÀ.

1) BISOGNA DECIDERE CHE VALORE DI TENSIONE HA V_D, V_S CON $V_D = 6V$, $\Rightarrow V_S = 2V$
$R_S = 20\ K\Omega$; $R_D = 60\ K\Omega$

2) BASTA CHE $R_1 = R_2$
$R_1 = R_2 = 1\ M\Omega$